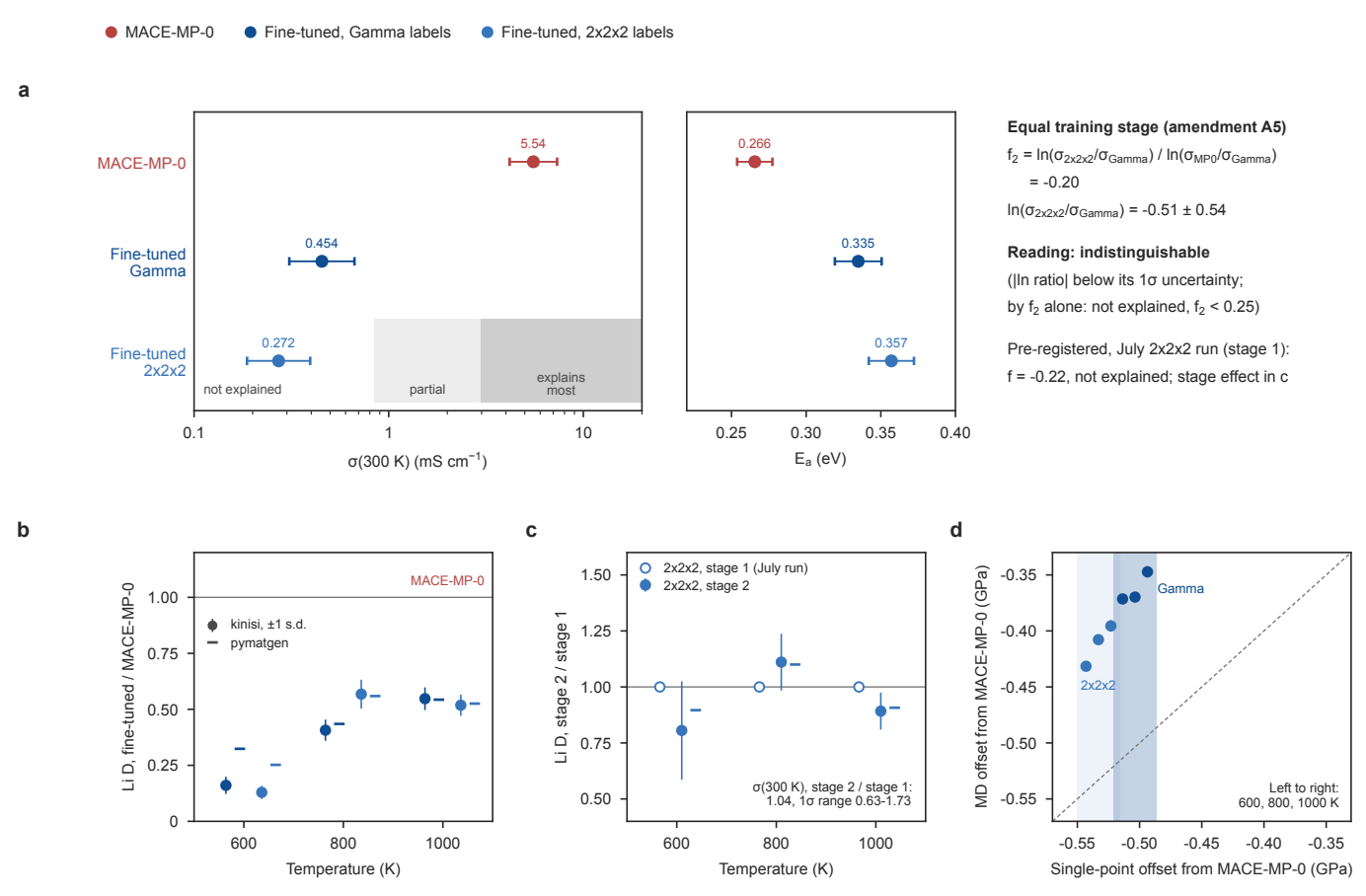

MACE-MP-0
Fine-tuned, Gamma labels
Fine-tuned, 2x2x2 labels
a
MACE-MP-0
Fine-tuned Gamma
Fine-tuned 2x2x2
5.54
0.454
0.272
not explained
partial
explains most
0.1
1
10
σ(300 K) (mS cm−1)
0.266
0.335
0.357
0.25
0.30
0.35
0.40
Ea (eV)
Equal training stage (amendment A5)
f2 = ln(σ2x2x2/σGamma) / ln(σMP0/σGamma)
= -0.20
ln(σ2x2x2/σGamma) = -0.51 ± 0.54
Reading: indistinguishable
(|ln ratio| below its 1σ uncertainty;
by f2 alone: not explained, f2 < 0.25)
Pre-registered, July 2x2x2 run (stage 1):
f = -0.22, not explained; stage effect in c
b
MACE-MP-0
kinisi, ±1 s.d.
pymatgen
Li D, fine-tuned / MACE-MP-0
1.00
0.75
0.50
0.25
0
600
800
1000
Temperature (K)
c
2x2x2, stage 1 (July run)
2x2x2, stage 2
Li D, stage 2 / stage 1
1.50
1.25
1.00
0.75
0.50
σ(300 K), stage 2 / stage 1:
1.04, 1σ range 0.63-1.73
600
800
1000
Temperature (K)
d
Gamma
2x2x2
MD offset from MACE-MP-0 (GPa)
-0.35
-0.40
-0.45
-0.50
-0.55
Left to right:
600, 800, 1000 K
-0.55
-0.50
-0.45
-0.40
-0.35
Single-point offset from MACE-MP-0 (GPa)